AUTHOR TESTED VIDEO PROOF PROVIDED

POCKET-SIZED THERMAL CAMERA
Using AMG8833

ASHWINI KUMAR SINHA

Thermal cameras are used for PCB fault detection, solar-panel inspection, engineering thermal analysis, and body-temperature monitoring. However, good-quality thermal cameras are often expensive and relatively bulky.

This system provides an affordable, highly portable option for basic thermal imaging and analysis. It uses the AMG8833 8 × 8 infrared sensor to capture thermal data and can be upgraded to higher resolution by replacing it with an MLX90640 or MLX90641. A GC9A01 round display provides real-time thermal video with colour mapping, making temperature variations easy to view.

Measuring just 3.5cm in diameter, the compact, battery-powered unit is small enough to carry like a keychain. It can measure the temperature of a person or object and help identify abnormal hotspots in electronic devices, overloaded components, poor solder joints, and other sources of excessive heat. It can also be used for basic PCB thermal analysis and routine thermal checks.

Fig. 1 shows the author's prototype, including the rear and top views, along with a size comparison against a scale. The components required to build this system are listed in the Bill of Materials table.

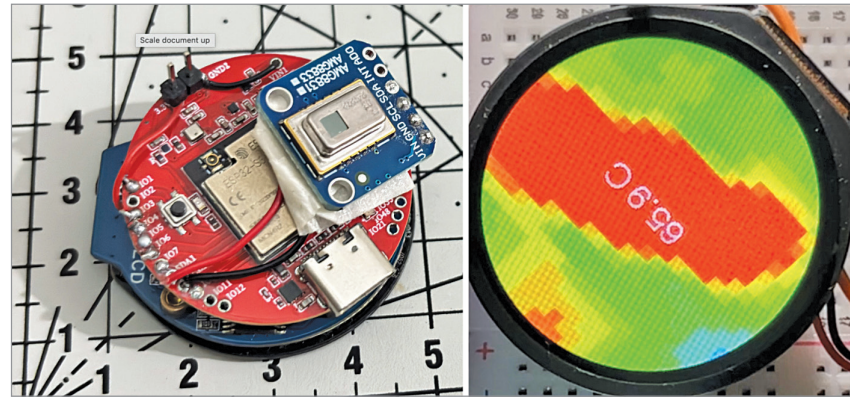


Fig. 1: Author's prototype with backside and top views, along with a size comparison on a scale

BILL OF MATERIALS

Name	Designator	Description	Quantity
IPS/LCD display	U1	GC9A01 IPS display	1
ESP32-S2 board	U2	IndusBoard COIN V2 or any ESP32-S2-based board	1
Thermal sensor	U3	AMG8833 thermal sensor module	1
3.3V power supply	—	3.3V Li-Po battery, 800mAh	1

Circuit and working

Fig. 2 shows the circuit diagram of the thermal camera based on the IndusBoard Coin V2. The circuit consists of four main components: the GC9A01 round IPS display (U1), the IndusBoard Coin V2, an ESP32-S2-based coin-sized board (U2), and the GY-AMG8833 thermal sensor (U3).

The GY-AMG8833 thermal sensor (U3) contains an 8 × 8 infrared array that measures temperature at 64 points simultaneously. It communicates with the IndusBoard Coin V2 through the I²C interface, with SDA and SCL connected to GPIO 8 and GPIO 9, respectively. The sensor's VIN is connected to the 3.3V supply, while GND is connected to the common ground. The INT and AD0 pins are left unconnected in the basic configuration.

The IndusBoard reads the 64 temperature values from the AMG8833 and processes them to generate a thermal image. The temperature values are mapped to different colours according to their levels, producing a thermal-image effect. The processed data is then sent to the GC9A01 round TFT display (U1) through the SPI interface using the SDI (MOSI), SCK, CS, and D/C signals. The recommended SPI connections shown in the schematic are Pin 1 to CS, Pin 3 to SCK, Pin 4 to SDI (MOSI), and Pin 5 to D/C.

The GC9A01 display presents the resulting thermal image in real time. Its LED+ pin is connected to VCC and LED− to GND for backlight operation. The display communicates with the IndusBoard through the SPI interface using the assigned GPIO pins.

Power is supplied through common 3.3V (VCC) and GND lines shared by all three modules. Thus, the overall signal flow is AMG8833 thermal sensing→I²C→IndusBoard processing→SPI→GC9A01 thermal display. The IndusBoard can also